

COL 703/7203

31st August

We assume that we are working with
a countable set of propositions.

$\{p_1, p_2, \dots\}$

$\neg$ $\vee$

$\phi ::=$ $p \mid \neg \psi \mid \psi_1 \vee \psi_2$

Derived connectives

$$\phi_1 \wedge \phi_2 = \neg(\neg\phi_1 \vee \neg\phi_2)$$

$$\phi_1 \rightarrow \phi_2 = \neg\phi_1 \vee \phi_2$$

$$\phi \equiv \psi = (\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)$$

Let Φ be the set of all propositional
logic formulas.
 $\hookrightarrow$ a countably infinite set.

For all formulas ϕ , if ϕ is consistent then ϕ is satisfiable.

consistent formula(ϕ) $\rightarrow$ maximal consistent set of formulas (X)

the canonical valuation makes all formulas in X true.



canonical valuation (v_X)

v_X

$\phi \mapsto T$ if $\phi \in X$
 $\phi \mapsto F$ otherwise

~~prop~~ i) ϕ is consistent if $\not\vdash \neg \phi$

ii) a finite set of formulas $\{\phi_1, \dots, \phi_n\}$ is consistent if $\bigwedge_{i=1}^n \phi_i$ is consistent.

iii) an arbitrary set of formulas is consistent if every finite subset is consistent.

Claim 1

Every consistent set can be extended to an MCS.

Let X be an arbitrary consistent set.

$X_0, X_1, \dots$

— $X_0 = X$

—
$$X_i = \begin{cases} X_{i-1} \cup \{\phi_i\} & \text{if } X_{i-1} \cup \{\phi_i\} \\ & \text{is consistent} \\ X_{i-1} & \text{otherwise} \end{cases}$$

Each X_i is consistent

$$X_0 \subseteq X_1 \subseteq X_2 \subseteq X_3 \subseteq \dots$$

Let
$$Y = \bigcup_{i \geq 0} X_i$$

We claim that Y is a maximal consistent set.

To prove :

- i) Y is consistent
- ii) Y is maximal

i) If γ is not consistent

then $\{\psi_1, \dots, \psi_n\} \subseteq \text{for } \gamma$
that is not consistent.

$\parallel$
 ϕ_j

(let ϕ_j last
formula in
the
enumeration
of
formulas
just
was
added
(to X_j)

then this $\{\psi_1 \dots \psi_n\}$
is also a finite
subset of X_j
which means that X_j
is inconsistent -
contradiction!

X_j

X_{j+1}	X_{j+2}
ϕ_{j+1}	ϕ_{j+2}

ii) To prove $\boxed{\gamma \text{ is maximal}}$ There is a $\phi_k \notin \gamma$
~~Suppose not.~~

such that $\gamma \cup \{\phi_k\}$ is consistent.

but ϕ_k was not added.
 Therefore, there must be a finite subset Z of γ such that $Z \cup \{\phi_k\}$ is inconsistent.

$$X_k = \boxed{X_{k-1} \cup \{\phi_k\}}$$

if the whole set is consistent

otherwise

$$X_{k-1} \cup \{\phi_k\} \Rightarrow X_{k-1}$$

But Z is also a finite inconsistent subset of $\gamma \cup \{\phi_k\}$.
 Therefore, $\gamma \cup \{\phi_k\}$ cannot be consistent.

Claim 2 Let X be an MCS. Then

i) for all formulas ϕ , $\phi \in X$ iff $\neg \phi \notin X$

ii) for all formulas ϕ, ψ ,
 $\phi \vee \psi \in X$ iff $\phi \in X$
or $\psi \in X$.

Claim

Let X be an MCS. Then for all

$$\phi \in X \text{ iff } v_x \models \phi.$$

Proof by induction on structure of ϕ .

ϕ is an atom. (Trivial)

Base case

Inductive step

$$\phi = \neg \psi$$

$$v_x \models \neg \psi$$

iff

$$v_x \not\models \psi$$

(from the property of valuation)

iff

$$\psi \notin X$$

(induction hypothesis)

iff

$$\neg \psi \in X$$

(from claim 2)

$$\gamma_x \models \phi \vee \psi$$

iff

$$\gamma_x \models \phi \text{ or } \gamma_x \models \psi$$

iff

$$\phi \in X \text{ or } \psi \in X$$

(induction hypothesis)

iff

$$\phi \vee \psi \in X \text{ (claim 2)}$$

Claim 2 (proof). Let X be an MCS. Then

i) for all formulas ϕ , $\phi \in X$ iff $\neg\phi \notin X$.

→ both ϕ and $\neg\phi$ cannot be in X

→ at least one of them must be in X .

Suppose $\{\phi, \neg\phi\} \subseteq X$.

because $\{\phi, \neg\phi\}$ must be consistent

$\nVdash \neg(\phi \wedge \neg\phi)$

$\nVdash \neg(\neg(\neg\phi \vee \neg\neg\phi))$

$\nVdash \neg(\neg\phi \vee \neg\neg\phi)$

$\nVdash (\neg\phi \vee \phi)$

$\nVdash (\phi \rightarrow \phi)$

$\nVdash \phi \rightarrow \phi$

but this is a contradiction because $\phi \rightarrow \phi$ is derivable.

Since $\neg\neg\psi$ and ψ are equivalent

We know that: $\neg\neg\phi \rightarrow \phi$ (last class)

To prove: $\phi \rightarrow \neg\neg\phi$

$\neg\neg\neg\psi \rightarrow \neg\psi$ (substituting $\neg\psi$ for ϕ in this)

$(\neg\neg\neg\psi \rightarrow \neg\psi) \rightarrow (\psi \rightarrow \neg\neg\psi)$

we had proved
earlier that
 $(\neg\psi \rightarrow \neg\phi) \rightarrow (\phi \rightarrow \psi)$

MP gives us $(\psi \rightarrow \neg\neg\psi)$.

But we claim that at least one of $\phi, \neg\phi$ must be in X .

We'll assume that both are not there in X and then arrive at a contradiction.

Next class